

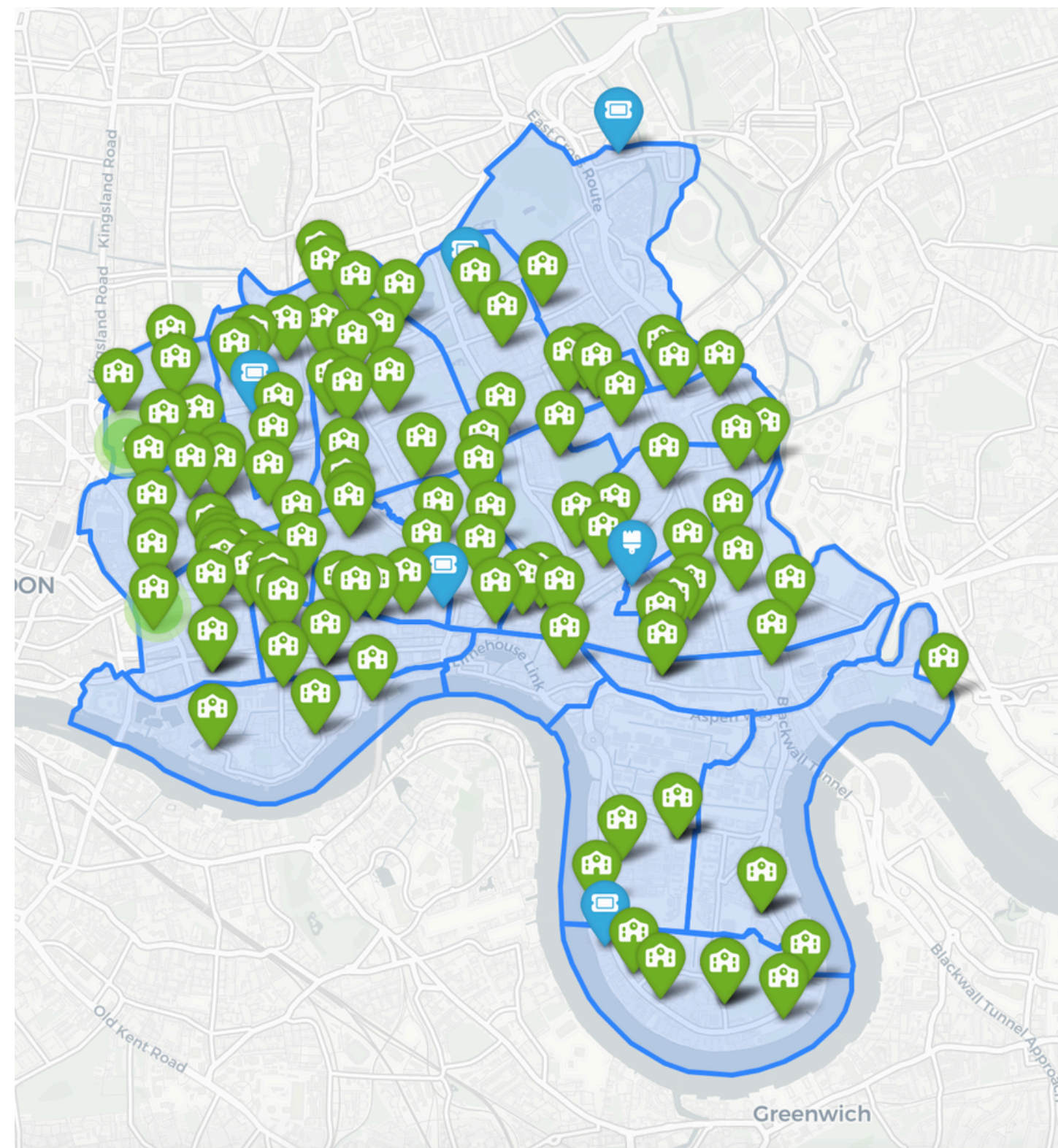


Mapping Cultural Infrastructure and Educational Institutions in Tower Hamlets



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Problem Description

- **Local Issue:** Despite London's reputation for education and culture, many boroughs like Tower Hamlets are often overlooked.
- **Importance:** Access to quality schools and cultural amenities significantly impacts community well-being and living choices.
- **Project Objective:** To reveal and visualize the abundance of educational institutions and cultural facilities in Tower Hamlets, helping residents make informed decisions on where to live or work.



Datasets Overview



- School Data: Information on schools (names, types, locations) from the London School Atlas.
- Quality Ratings: Ofsted ratings data providing insight into school performance.
- Geographical Data: Boundaries and ward information for Tower Hamlets.
- Cultural Infrastructure: Details on cinemas, libraries, museums, theatres, etc., provided by the Greater London Authority.
- Why It Works: These sources offer comprehensive and reliable data that together paint a full picture of the borough's educational and cultural offerings.

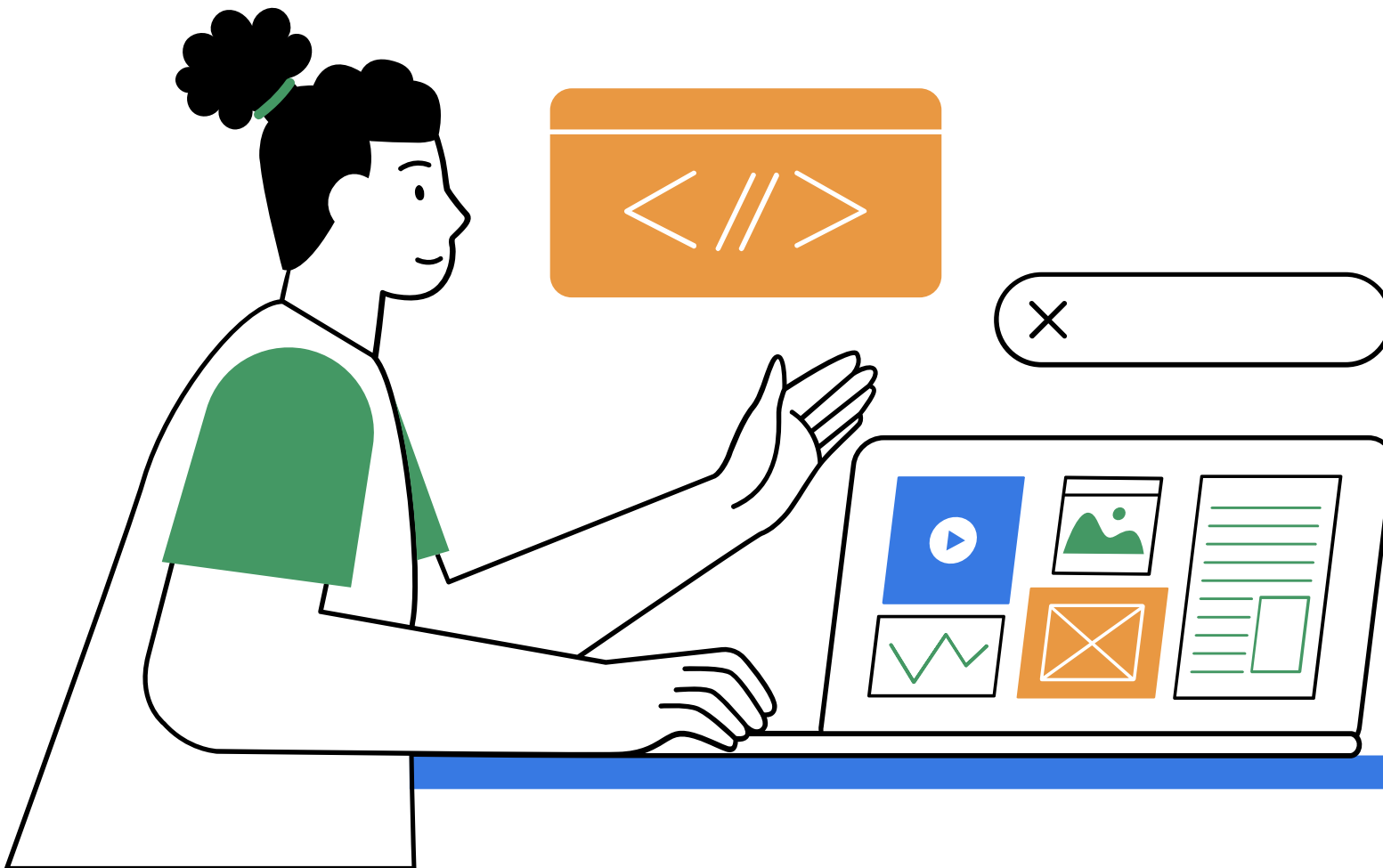
SCHOOL_NAM	TYPE	PHASE	GENDER	WARD_NAME	WEBLINK	URN	POINT_X	POINT_Y
Ben Jonson Primary School	Community School	Primary	Mixed	St. Dunstan's and Stepney Green	www.benjonson.towerhamlets.sch.uk	100890	-0.038560	51.521041
Bonner Primary School	Community School	Primary	Mixed	Mile End and Globe Town		100891	-0.048668	51.529725
Old Palace Primary School	Community School	Primary	Mixed	Bromley-by-Bow	http://www.oldpalaceprimary.co.uk/	100892	-0.014142	51.527636
Canon Barnett Primary School	Community School	Primary	Mixed	Spitalfields and Banglatown		100893	-0.071784	51.516322
Cayley Primary School	Community School	Primary	Mixed	St. Dunstan's and Stepney Green		100894	-0.038386	51.516047



Data Preparation

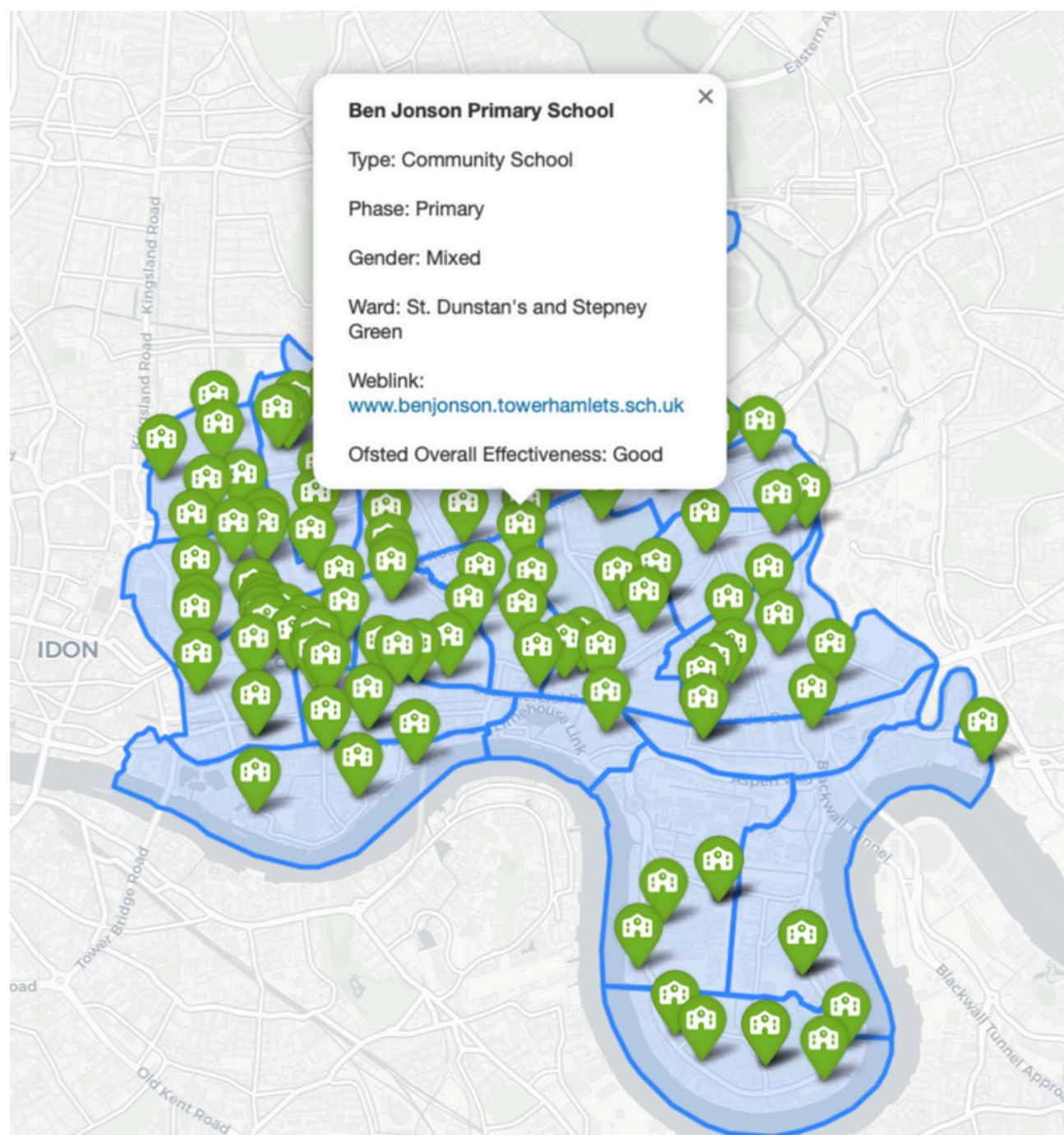


- Cleaning: Removed duplicate entries and ensured all location information was complete.
- Refinement: Handled missing values and standardized data for consistency.
- Merging: Combined school information with quality ratings using unique reference numbers.
- Result: A clean, integrated dataset ready to map schools and cultural amenities accurately.



```
.. counts_tower_hamlets_schools_data = tower_hamlets_schools_data['SCHOOL_NAM'].value_counts()  
   duplicate_counts = counts_tower_hamlets_schools_data[counts_tower_hamlets_schools_data > 1]  
   print(duplicate_counts)
```

```
SCHOOL_NAM  
Phoenix School                2  
St Paul's Way Trust School    2  
River House Montessori School 2  
Al-Mizan School              2  
Excellence Christian School    2  
Name: count, dtype: int64
```



- Mapping: Plotted schools and cultural venues on a map of Tower Hamlets.
- Distribution Insights: Identified how schools and cultural amenities are spread across different wards.
- Quality Overview: Visualized Ofsted ratings to highlight areas with high-performing schools.
- Takeaway: The visuals reveal patterns and hotspots, guiding where community resources are concentrated.



Exploratory Data Analysis (EDA)



Model Selection

Model 1 Grouping Cultural Amenities

- The algorithm divides cultural spots into a set number of groups.
- It then calculates a central point for each group—this represents the ideal spot to be near many cultural venues.



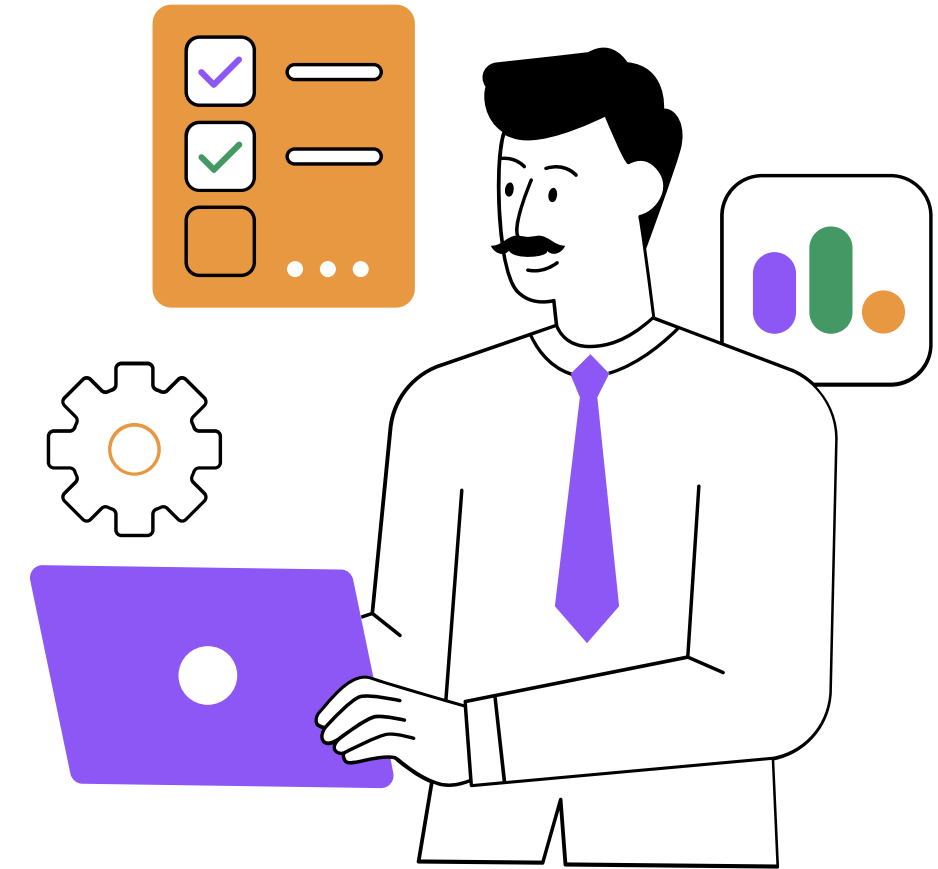
Model 2 Finding Nearest Amenties

- A simple method is used to determine the 5 closest cultural facilities to any selected school.



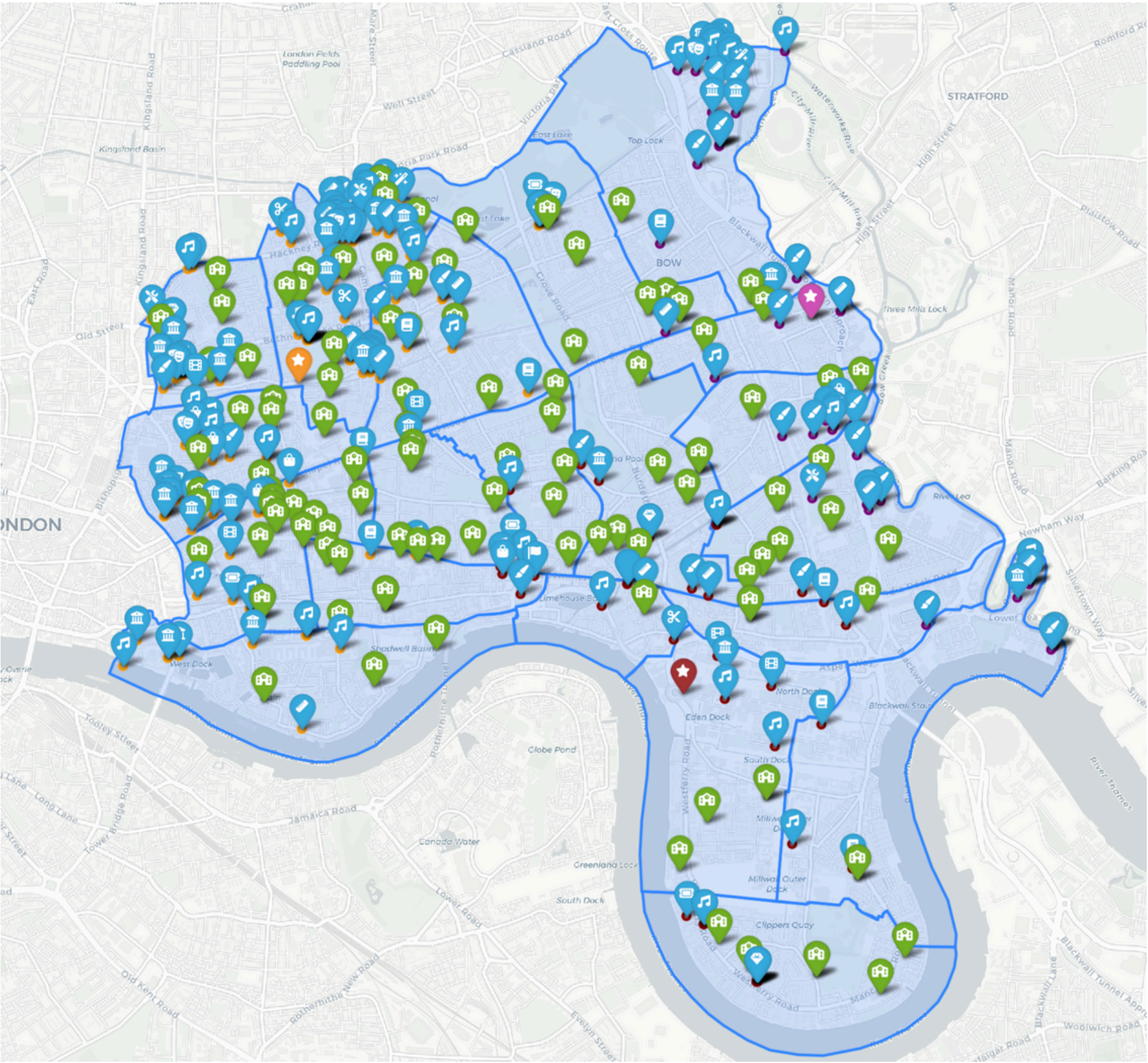
Purpose

Both approaches provide intuitive visual insights without overwhelming technical details.





Model 1 Grouping Cultural Amenities



Model 2 Finding Nearest Amenties

5 Closest Cultural Attractions to Ben Jonson Primary School			
	name	category	distance
0	Local History Library	Libraries	0.0032
1	Copperfield Road	Artists workspaces	0.0034
2	Haileybury Centre	Dance rehearsal studios	0.0052
3	Ragged School Museum	Museums and public galleries	0.0053
4	Half Moon Young People's Theatre	Theatre	0.0077



Model Evaluation



Clustering Results

- Clearly shows cultural “hotspots” across Tower Hamlets.
- Note: The suggested centre points are ideal locations, but real-life obstacles (e.g., roads or buildings) aren’t considered.

Distance Calculations

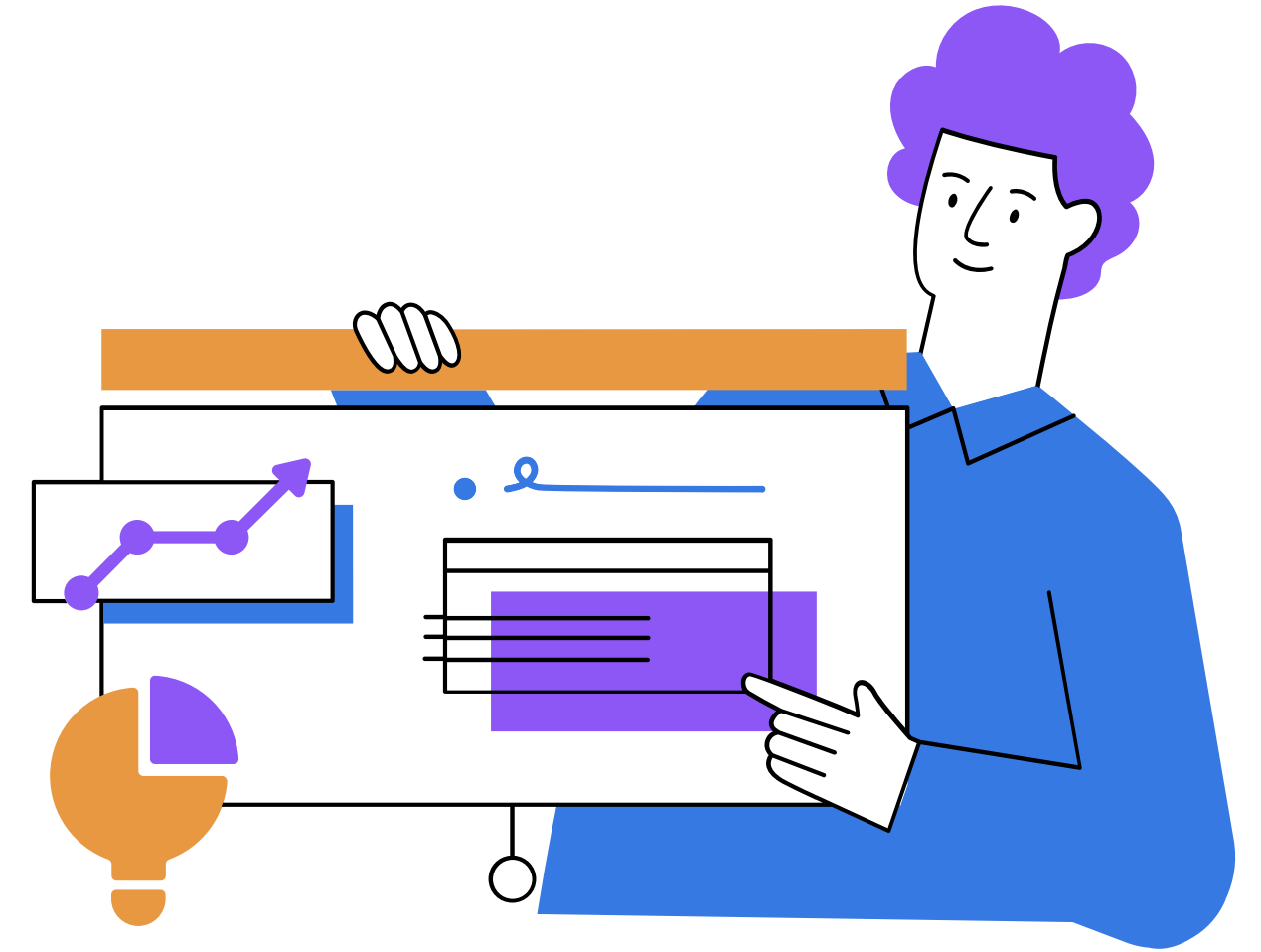
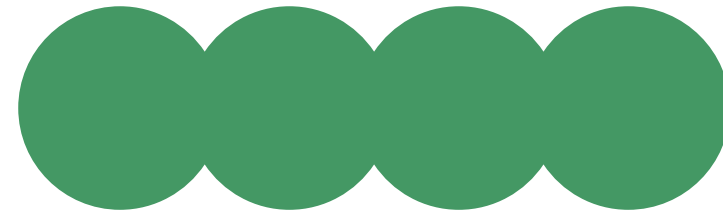
- Helps identify nearby cultural spots for a given school.
- Reminder: The straight-line distances may differ from actual travel distances.

Overall Impact

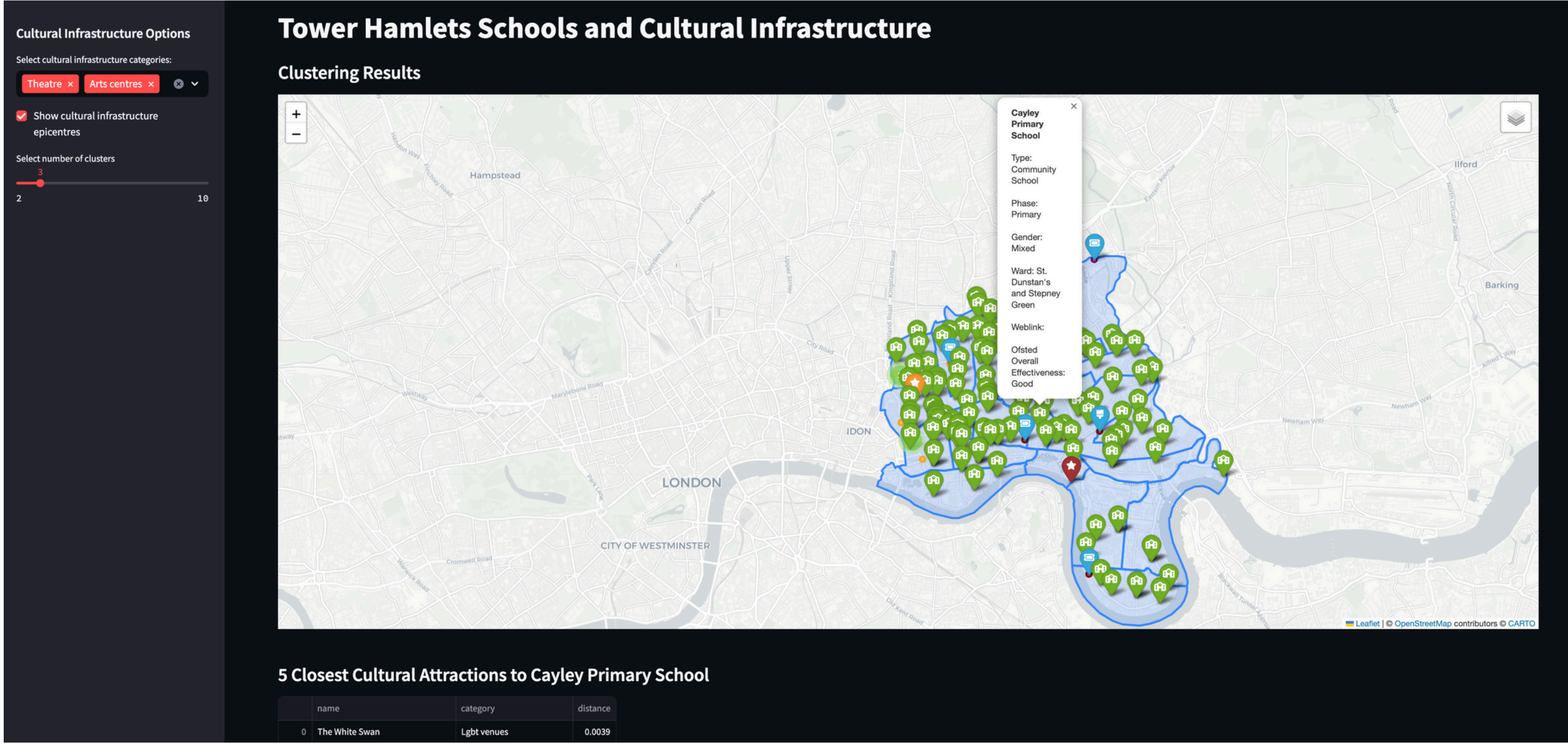
The models simplify complex data into actionable insights for the community.



Software Deployment



- Interactive Application:
 - Developed using Streamlit, the app presents an interactive map of Tower Hamlets.
 - Users can explore schools, cultural venues, and identified hotspots.
- User Experience:
 - Options to select cultural categories and the number of hotspots.
 - On clicking a school, the app shows the 5 nearest cultural facilities.
- Benefits: Empowers local residents, families, and stakeholders with easy-to-understand visual data to make informed decisions.





Cultural Infrastructure Options

Select cultural infrastructure categories:

Libraries

Museums and p...

Theatre

Arts centres

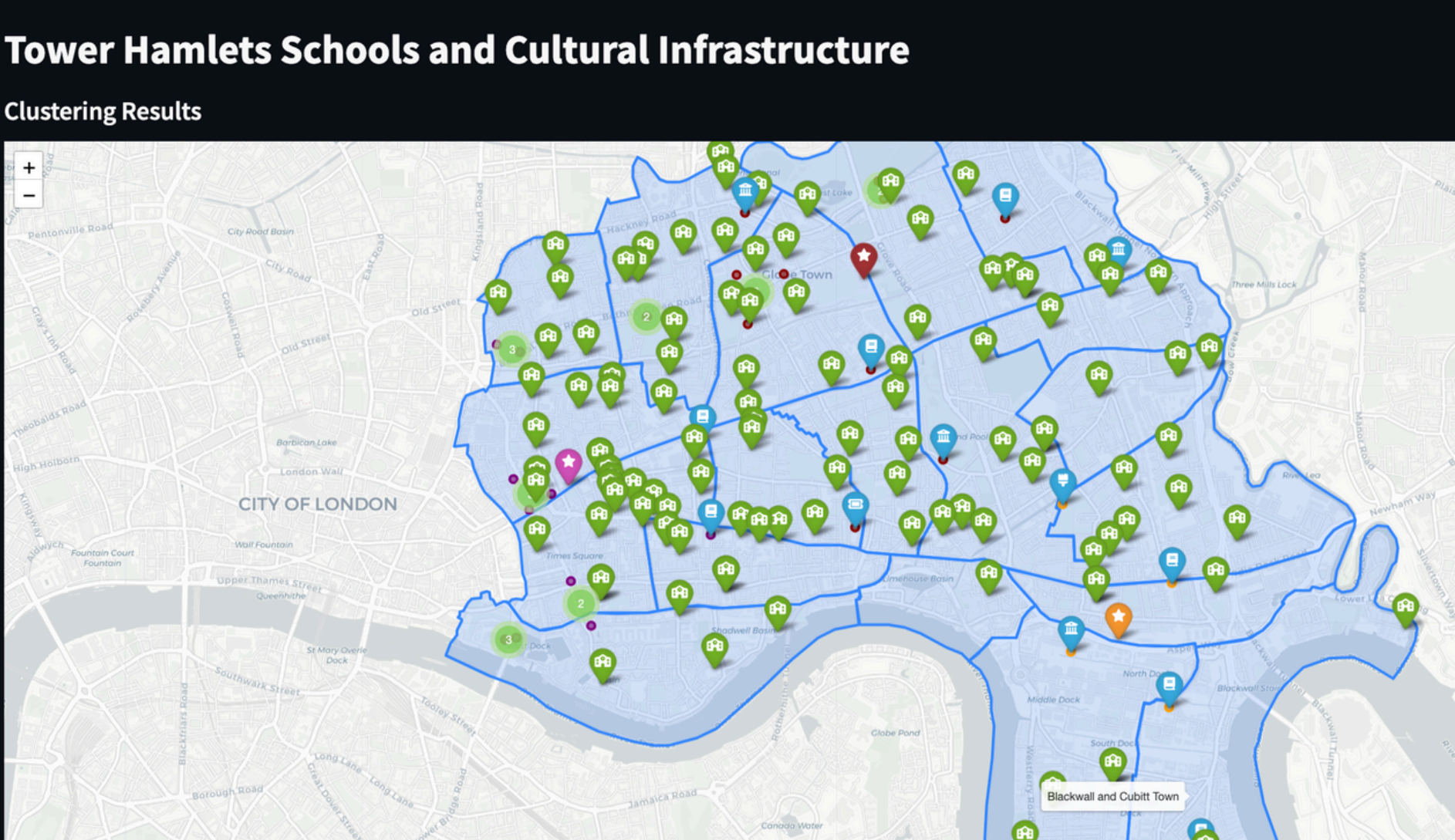
☒ Show cultural infrastructure epicentres

Select number of clusters

3

2

10



Cultural Infrastructure Options

Select cultural infrastructure categories:

Libraries

Museums and p...

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Arts centres

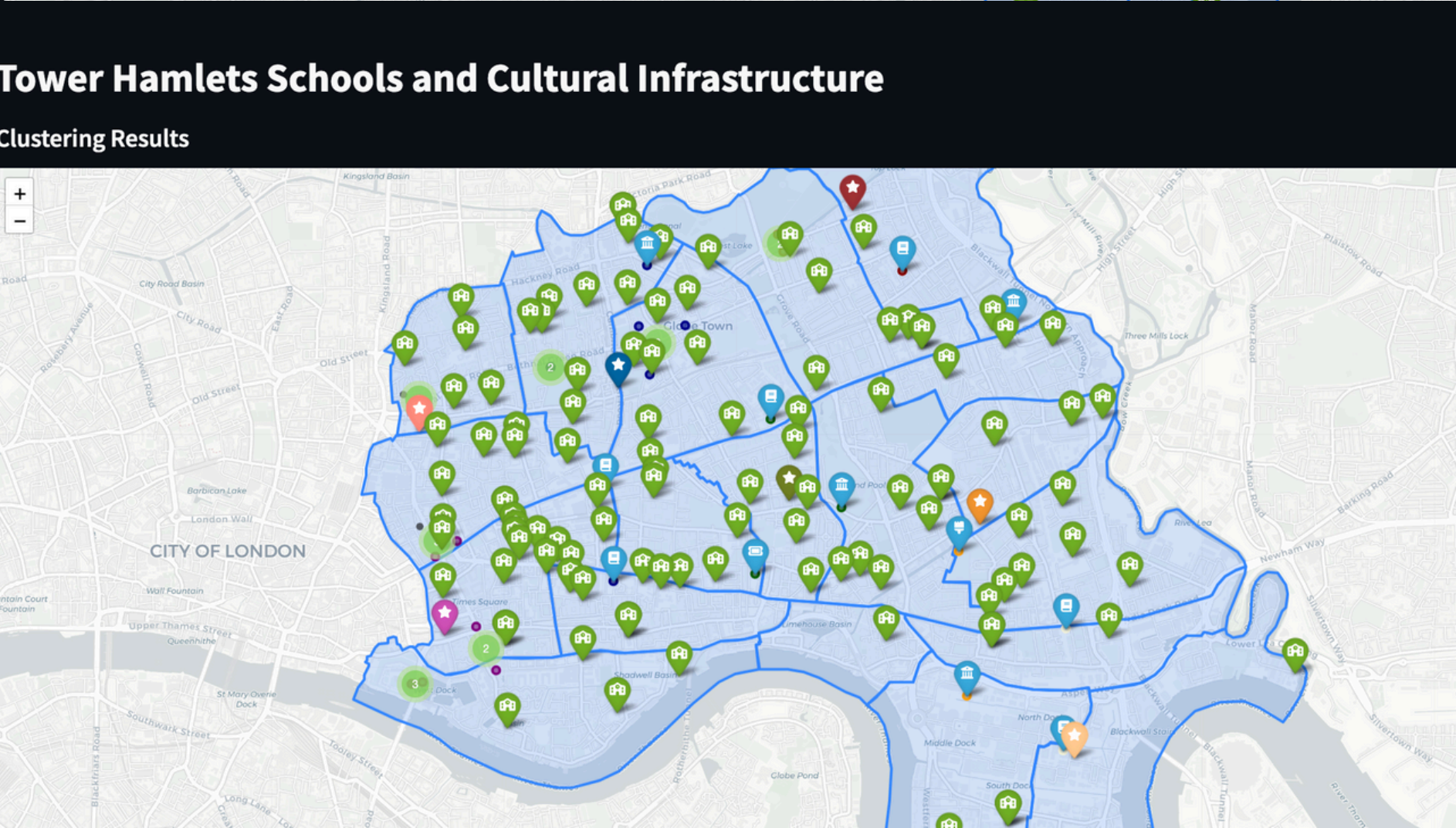
☒ Show cultural infrastructure epicentres

Select number of clusters

8

2

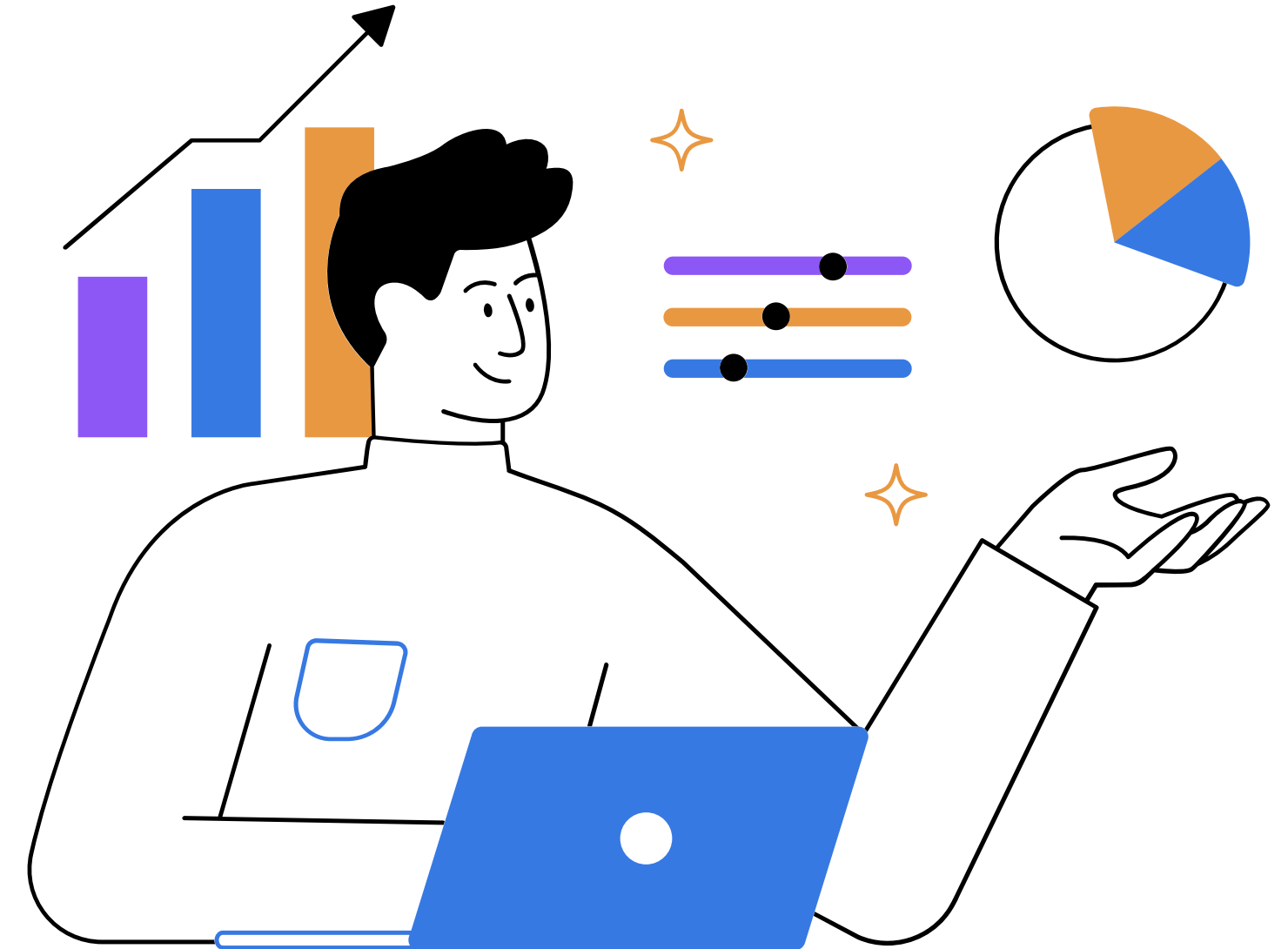
10



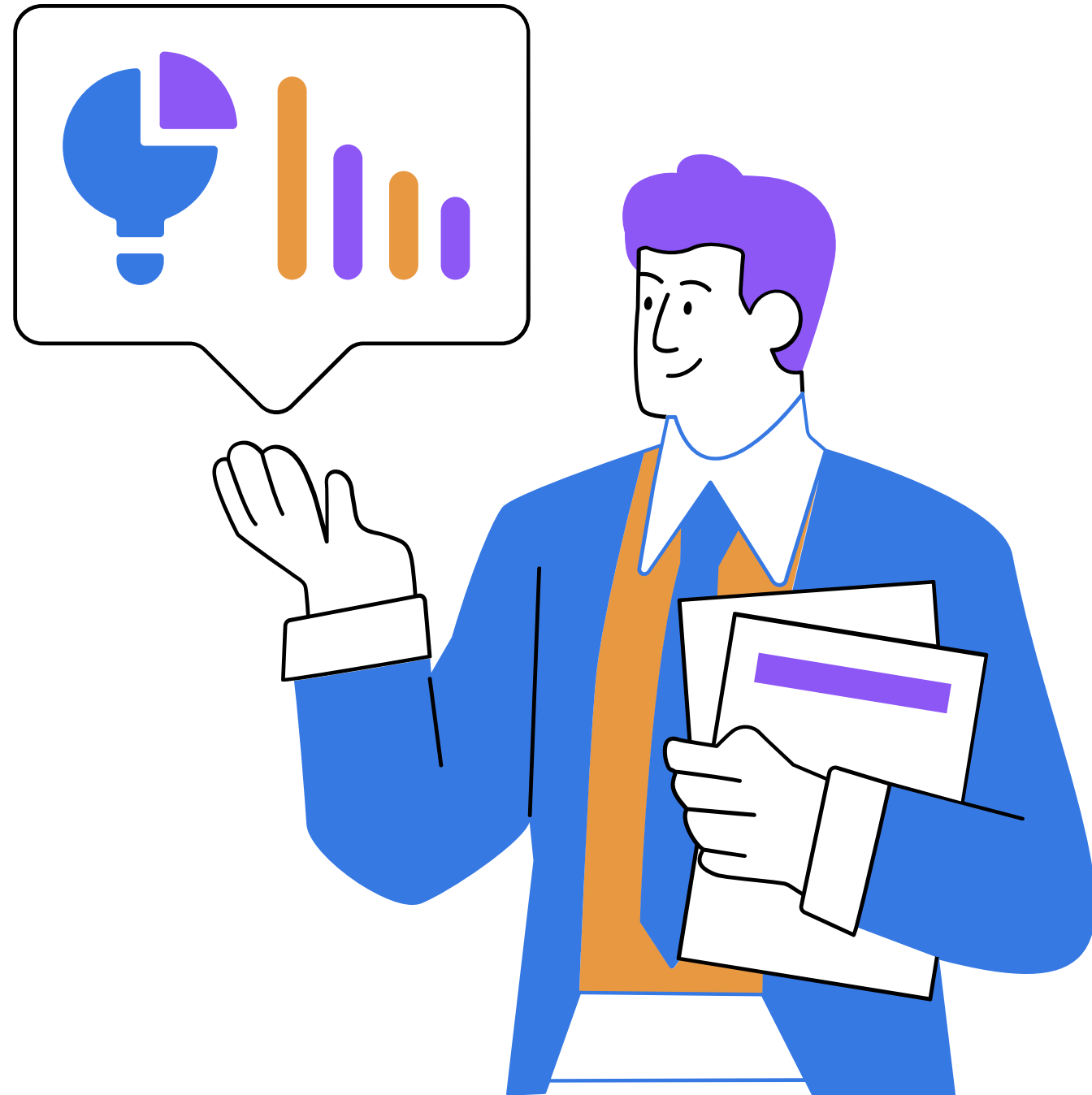


- Data Challenges:
 - Some schools lack quality ratings and duplicate entries had to be managed.
- Modelling Limitations:
 - The grouping and distance methods use simplified calculations and do not consider real-world travel constraints.

Challenges and Limitations



- Ethical Considerations:
 - Ensuring accurate, fair, and non-biased presentation of public data.
- These challenges offer valuable lessons and directions for further improvement.



Conclusion & Future Development

- Key Findings:
 - Comprehensive mapping of schools and cultural venues in Tower Hamlets.
 - Identification of cultural hotspots and accessible school locations.
- Societal Impact:
 - Supports residents in choosing locations that best suit their lifestyle and access needs.
 - Provides insights that could influence community planning and investment.
- Next Steps:
 - Expand the analysis to include additional boroughs.
 - Incorporate extra factors (e.g., crime, income, green spaces) for a holistic view.
 - Enhance the application with real-world travel data and community feedback.



Thank You

